

Nested residual classes refute Conjecture 00000000388

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Statement. The source defines, for positive integers m ,

$$W_m = \{\xi : w_n(\xi) \leq n/m \text{ for infinitely many } n, \quad \xi \notin U_k \text{ for every } k\}.$$

It asserts that each W_m is nonempty and uncountable, that distinct W_m are disjoint, and that their union is the set of transcendental numbers. The nonemptiness and disjointness assertions are already incompatible with this definition.

Proof. Let $1 \leq m \leq k$. For every nonnegative integer n ,

$$\frac{n}{k} \leq \frac{n}{m}.$$

Consequently, if $w_n(\xi) \leq n/k$, then $w_n(\xi) \leq n/m$. The same infinitely many indices witnessing the condition for W_k therefore witness the condition for W_m . The exclusion from all U_j is identical in both definitions. It follows that

$$W_k \subseteq W_m \quad (1 \leq m \leq k).$$

In particular $W_2 \subseteq W_1$. If the asserted nonemptiness holds, choose $\xi \in W_2$; then $\xi \in W_1 \cap W_2$, contradicting the asserted disjointness for $1 \neq 2$. Equivalently, disjointness forces $W_2 = \emptyset$. This refutes the conjunction in the source, regardless of its further uncountability and coverage assertions. $\square$

Scope. The argument does not assert the existence of any particular approximation orders or transcendental numbers with these properties. It proves that no assignment of ordered numerical values to the symbols $w_n(\xi)$ can make the stated threshold definition and both assertions hold simultaneously. It also applies if the values can be $+\infty$, or if the underlying set is restricted to transcendental numbers. No Diophantine approximation theorem is needed.

Formal verification. The accompanying Lean 4.19.0 project uses only `Std`. It proves the following statement for an arbitrary underlying type X , arbitrary approximation-order data, and an arbitrary family of excluded sets U_k :

$$\neg[(\forall m \geq 1, W_m \neq \emptyset) \wedge (W_m \text{ are pairwise disjoint})].$$

Numerical representation and semantic bridge. `RationalBound` stores a numerator $a \in \mathbb{N}$ and a positive denominator b , representing the nonnegative rational number a/b . `BoundLE` compares two such fractions by cross multiplication: $a/b \leq c/d$ means $ad \leq cb$. Thus this representation uses exact rational bounds, not integer division. The theorem `threshold_antitone` proves $n/k \leq n/m$ for every n and every $1 \leq m \leq k$ using monotonicity of multiplication in $\mathbb{N}$.

An approximation order is represented by `UpperCut`, the set of its rational upper bounds. For an ordinary numerical value v , this is $\{q \in \mathbb{Q}_{\geq 0} : v \leq q\}$; it is upward closed, also when $v = +\infty$ (in which case it is empty). The formal theorem permits every upward-closed cut, a larger class than real-valued approximation orders, so its universal impossibility conclusion applies in particular to the usual numerical values. `UpperCut.respects_equivalent` proves independence of the chosen fraction representation.

To make the interpretation explicit, `OrderedValues` records any numerical comparison relation with transitivity and an order-preserving interpretation of rational bounds. `cutOfValue` constructs the corresponding upper cut; `cutOfValue_exact` verifies that membership is precisely the comparison $v \leq q$. The final theorem `ordered_value_version` states the contradiction directly using those numerical comparisons. The project does not construct real numbers or the analytic definition of w_n ; neither is required, because the theorem is universal in the values and uses only the indicated order laws.

Infinitude and source quantifiers. `InfinitelyOften p` means $\forall B \in \mathbb{N}, \exists n > B, p(n)$. For a subset of $\mathbb{N}$ this is exactly infinitude. Passing to a weaker inequality keeps each witness n , as verified by `InfinitelyOften.mono`. `NoU` quantifies over every positive integer index, and `W` uses the same family of excluded sets for every m . The theorem `W_antitone` proves the full nestedness statement; `conjecture388_refuted` then negates the two necessary clauses of the original conjecture. Both this theorem and `ordered_value_version` have no axiom dependencies. There are no added axioms, proof placeholders, or native decision procedures. The project treats warnings as errors.